

Chapter 108 — Urinary Incontinence and Pelvic Prolapse: Epidemiology and Pathophysiology

Gary E. Lemack and Maude Carmel · Campbell-Walsh-Wein Urology, pp. 2218–2237

DEFINITION AND CLASSIFICATION OF URINARY INCONTINENCE

Introduction and Overview of the Lower Urinary Tract

- **Urinary incontinence (UI)** affects millions of women and men and their families, at a cost of **billions in direct and indirect healthcare spending**.
- **UI is heavily under-reported**. Visits and procedures have risen for decades, yet many symptomatic women are undiagnosed. In one managed-care population, **over half of women with no UI recorded at their visit reported UI in the past year when asked directly**, and over half of them graded it moderate to severe. Primary care must **ask directly** and refer.
- The **lower urinary tract (LUT)** is the **bladder and urethra**, with complex neural innervation and musculofascial pelvic support.
- **Lower urinary tract symptoms (LUTS)**, including UI, arise from **anatomical abnormality or structural deterioration** of the LUT or its surroundings, at macroscopic, microscopic or ultrastructural level (e.g. urothelium, detrusor, sphincteric unit).
- **Two normal functions: storing urine at low intravesical pressure and expelling it at socially appropriate times**. Both are shaped by coexisting disease, ageing, demographic and environmental risk, lifestyle and genetics.

Signs, Symptoms, and Urodynamic Observations of Urinary Incontinence

- **LUTS are non-specific**: different causes of LUT dysfunction (LUTD) give similar symptoms. Urgency occurs in **overactive bladder (OAB)**, **bladder outlet obstruction (BOO)**, **pelvic organ prolapse (POP)** and **interstitial cystitis/painful bladder syndrome (IC/PBS)**.
- **Work-up is tailored** to the condition, the risk of the proposed intervention and clinician and patient confidence. A definitive diagnosis targets treatment better, but **extensive, expensive evaluation is not always needed first**.

SIGNS VS SYMPTOMS VS OBSERVATIONS

	Definition and nature	Examples
Sign	Observable by the clinician; reproducible and objective	UI on supine stress test; POP on examination in a woman reporting pelvic pressure
Symptom	Subjective report by the patient or a caregiver of a condition, or of a change from previous experience; descriptive and experiential	Self-report; validated questionnaires, which try to make a subjective report more objective
Observation	Non-urodynamic, or urodynamic (UD): findings during urodynamic studies (UDS) that show the physiological basis of LUTS	Non-UD: bladder diary. UD: storage (cystometrogram), voiding (pressure–flow) and post-micturition abnormalities

- **Having a symptom is not the same as being bothered by it. Quality-of-life (QoL) impact is the commonest reason to intervene** for LUTS. Without a change in QoL, treatment may not be indicated, and the same symptom may be very bothersome to one patient and go unnoticed by another.
- **Guidelines exist both for when to do UDS and for how to perform them**.

- **UDS aim to reproduce the patient's symptoms but often fail to.** When symptoms and UD observations disagree, **the clinician decides** what to do.

Concept — *Because symptoms are non-specific, LUTD is diagnosed by **triangulating three different kinds of evidence**: signs, symptoms and observations. The decision to treat is driven by **bother, not by the mere presence of a finding**.*

TERMINOLOGY OF LOWER URINARY TRACT SYMPTOMS AND INCONTINENCE

- Terminology for female LUTD is **standardised by a joint report of the International Continence Society (ICS) and the International Urogynecological Association (IUGA)** (2010), which builds on the 2002 ICS standardisation. The same documents also cover POP terminology and how to perform and interpret UDS.
- LUTS fall into **five groups: (1) UI, (2) bladder storage, (3) bladder sensation, (4) voiding and (5) post-micturition**.

Incontinence

- **UI is the complaint of involuntary loss of urine.**
- **Assessment must establish the precipitant** (urgency, a particular activity or action), **severity, impact on QoL, duration and frequency**.
- **No single tool answers all of these questions.** Questionnaires, bladder diary, examination and UDS are used together.

TABLE 108.1 LOWER URINARY TRACT SYMPTOM DEFINITIONS (REBUILT; ROWS MARKED † ARE DEFINED IN THE TEXT ONLY)

Symptom	Complaint of...	Sign, urodynamic correlate or distinction
Incontinence		
Stress UI (SUI)	Involuntary loss on physical exertion, sneezing or coughing, or other rise in intra-abdominal pressure	Sign: leakage per urethra synchronous with effort, Valsalva or cough. UD SUI: leakage on filling cystometry with a rise in abdominal pressure and no detrusor contraction
Urgency UI (UUI)	Involuntary loss associated with urgency	Occasionally seen on examination. UD: leakage with sudden urgency and no rise in abdominal pressure. No specific UD finding is required: detrusor overactivity (DO), oversensitivity and reduced capacity are all optional. DO incontinence (DOI) is a separate UD diagnosis
Mixed UI (MUI)	Involuntary loss with urgency and also with effort or raised intra-abdominal pressure	Urge-predominant, stress-predominant or equal
Postural UI †	Loss with a change of position	Typically sitting or lying → standing
Nocturnal enuresis	Involuntary loss during sleep	Not the same as nocturnal UUI , in which the patient wakes to void but cannot reach the toilet in time
Continuous UI	Continuous loss, day and night	Typically a LUT-vaginal fistula (vesicovaginal, ureterovaginal); few or no volitional voids
Insensible UI	Loss without awareness of how or when it happened	See Pathophysiology of Insensible Incontinence
Coital incontinence †	Loss with sexual intercourse	At penetration or intromission and/or at orgasm

Symptom	Complaint of...	Sign, urodynamic correlate or distinction
Storage and sensation		
Urinary (daytime) frequency	Voiding more often than previously deemed normal	≥8 voids/24 h is generally more than normal, but on its own does not imply dysfunction, because polyuria raises frequency
Urgency	Sudden, compelling desire to void that is difficult to defer	Pathological. Contrast with urge , the normal sensation that signals the need to void
OAB syndrome	Urgency, with or without UI, typically with frequency and nocturia	Defined by urgency
Nocturia	Interruption of sleep by the need to void	Each void is preceded and followed by sleep

- **What the table is for:** every entry is a **complaint**, so the terms are symptom-based. UD labels (UD SUI, DOI) are separate constructs that may or may not accompany them.
- **Traps:** UI does not need DO on UDS. UD SUI requires the absence of a detrusor contraction. Enuresis happens asleep; nocturnal UI happens after waking. Continuous leakage suggests a fistula.

Bladder Storage and Sensation

- Storage symptoms arise from abnormal bladder filling. Frequency, nocturia, urgency and OAB are defined in Table 108.1.

BLADDER SENSATION — ICS/IUGA CATEGORIES

Sensation	Definition and distinguishing point
Urge vs urgency	Urge is normal and signals the need to void. Urgency is sudden, compelling and difficult to defer
Increased	Desire to void comes earlier than previously normal, but voiding can be postponed , which is what separates it from urgency. On UDS the patient may report desire at <50 mL filling
Reduced	Desire comes later than previously, although the patient is aware of filling
Absent	Neither sensation of filling nor desire to void

Concept — *The terminology separates what the patient reports from what the test shows. Each term is a complaint, and a urodynamic label is a separate statement that neither proves nor excludes it.*

EPIDEMIOLOGY OF URINARY INCONTINENCE IN FEMALES

General Comments

- **Prevalence estimates vary widely, mainly because definitions and methods differ.** A single survey question in a mixed population, an incontinence-specific questionnaire in at-risk women, examination, UDS and diaries all give different figures. **Comparisons between studies, and before and after intervention, are therefore error-prone.** Societies have called for common outcomes and tools, but this is unresolved.
- **Questionnaire design drives the figure. The time frame matters most:** "any leak in the past month" gives a higher prevalence than a narrower window. **Bother-based definitions differ from presence-based ones**, and in a non-morbid condition bother should be paramount. Leakage shown on examination or UDS differs again from questionnaire report.
- **The population matters:** young nulliparous women and octogenarians in nursing facilities give very different figures. **Incidence and remission are even more bias-prone**, because prevalence is high relative to incidence and studies differ in interval, age, recall window and wording.

Pregnancy and the postpartum period

- **Pregnancy is probably the commonest cause of incident UI in young women**, and risk rises with gestational age. **Prevalence in pregnancy is ≈40%; SUI is the commonest type (about two-thirds).**
- **Risk factors for SUI in pregnancy:** obesity, pre-existing SUI, gestational diabetes, age, constipation, prior vaginal delivery, childhood enuresis.
- **Primiparous women have ≈3 times the risk of UI in pregnancy** compared with age-matched nulliparous women, still ≈2.5 times at 1 year. **Higher parity** probably adds risk up to 1 year postpartum (data mixed).
- **Postpartum remission cannot be assumed:** few women with UI at 3 months are dry at both 6 and 12 years.
- **Protective measures:** properly performed pelvic floor muscle training (PFMT) during and after pregnancy reduces incident UI in pregnancy and early postpartum, but **the benefit of antenatal exercise may be lost by 7 years**. **Postpartum weight loss** speeds recovery. **Supervised PFMT and electrical stimulation** treat postpartum UI effectively (systematic review). A validated model makes **vaginal delivery the strongest predictor of UI up to 8 weeks**.

Hysterectomy

- **Hysterectomy can both cause and resolve UI.** In a Swedish national registry, a minority of continent women developed de novo UI, predicted by **obesity and prior vaginal delivery**. Some with pre-existing UI were dry at 1 year, predicted by **uterine weight >300 g, body mass index (BMI) <25 kg/m² and coexisting POP**.
- **Vaginal and laparoscopic hysterectomy carry more UI than open hysterectomy** (10-year Finnish prospective study).

Incidence and remission

- **Women aged 40–60: annual incidence 1–10%, ≈3% per subtype.** Cohorts followed for 8–10 years show substantial cumulative new-onset UI.
- **SUI accounts for about half of new cases in younger cohorts.** SUI incidence, but not UUI or severe UI, rises across the menopausal transition (48–54 years). In older cohorts, annual incidence is ≈10–20%.
- **White women are more likely than Black women to develop UI over 5 years**, especially with higher baseline BMI or weight gain.
- **Annual remission is up to ≈12%.** It is higher in younger women and milder UI, and ≈30% with prolonged follow-up. **MUI remits least.** The highest remission is reported in Black women by some studies and, for SUI, in Hispanic women by others.
- **Remission implies transient causes**, as well as subjectivity and recall bias.

KEY POINTS: INCIDENCE AND REMISSION OF URINARY INCONTINENCE

- Pregnancy → incident UI in >40%. Hysterectomy resolves UI in 13%. Annual incidence 1–10% (≈3% per subtype). MUI remits least; **men remit considerably more often than women**.

Concept — Every UI prevalence figure depends on its definition, instrument, time frame and population. Stated numbers are therefore only comparable within a study. In young women, the event that dominates incidence is pregnancy.

RISK FACTORS FOR URINARY INCONTINENCE IN FEMALES

Aging

- **Ageing is a consistent independent risk factor**, and the risk keeps rising after the menopause. Over a third of Medicare beneficiaries (all ≥ 65) report UI, and the gradient with age persists even within this group.
- **With age, UI shifts away from SUI towards MUI and UUI.** In the Urologic Diseases in America study, overall prevalence was about half, with **SUI peaking at 40–59 and UUI/MUI predominating over 60.**
- **Over 75, UUI is linked to impaired mobility and SUI is not.** Better lower-limb function (strength, walking speed, "get up and go") predicts less incident UI over 4 years.
- **In nonagenarians, UI is independently linked to frailty and is an independent risk factor for death**, so treatment may be warranted even in the oldest patients.
- **Long-term care facility (LTCF) residence carries the greatest risk**, with prevalence $>40\%$ at admission. **Severe impairment of activities of daily living (ADL) is the strongest association.** Poor nutrition, immobility and dementia correlate with severity. Becoming incontinent while resident is driven by cognitive impairment, poor rehabilitation potential, stroke, Alzheimer disease and Parkinson disease.

Figure legends

- **Fig. 108.1** — UI subtypes by age in women (Aoki et al., *Nat Rev Dis Primers*, 2017): **SUI peaks in the early fifties then declines; urgency and mixed UI keep rising into the ninth decade.**
- **Fig. 108.2** — UUI (including MUI) by age in men and women (Milsom et al., *Eur Urol*, 2014): UUI rises with age in both sexes, and **in ageing men sometimes exceeds the female prevalence.**

Pregnancy

- **UI that develops in pregnancy raises the risk of future and persistent UI.** Prevalence falls by 3 months, but **$>30\%$ still have UI at 1 year, mostly SUI.** Nearly two-thirds of women with UI at 3 months still report it at 6 and 12 years.
- **Persistence is associated with older age at pregnancy and obesity. First pregnancy after age 30 roughly triples the odds of later UI** (systematic review). Alcohol and depression predict persistence; **physical activity and Black race protect.**

Aspects of Delivery

- **Caesarean section (CS) protects against later UI and SUI compared with vaginal delivery. Elective CS carries less risk than either CS after labour or vaginal delivery.** Primiparous women delivered by CS have less UI at 5 years.
- **Vaginal delivery leads to later SUI, not UUI. CS protects only if every delivery is by CS;** one vaginal delivery removes the benefit.
- **Number needed to treat (NNT) with CS to prevent one case of moderate-to-severe UI = 13.**
- **Unproven:** length of labour, forceps, anaesthesia, head circumference and episiotomy. There is **no conclusive evidence to change obstetric practice.** A long second stage is linked to immediate postpartum UI only.
- **Consistent: vaginal birth of a large baby.** At least one child ≥ 4 kg gives an odds ratio (OR) of ≈ 1.5 for weekly UI later in life.

Parity

- **A single pregnancy and delivery raises later UI risk (OR ≈ 1.5).** The effect of further deliveries is unclear and the data conflict; a meta-analysis shows a **modest rise with each delivery** (relative risk (RR) $\approx 1.4 \rightarrow 1.6$ from one to three).

- **Age at first birth is disputed:** most studies put the greatest risk at a first birth near age 20, whereas the systematic review above found risk raised by a first pregnancy after 30.
- **The parity association is strongest for SUI.**

Demographic Factors: Race, Ethnicity, Socioeconomics, and Geography

- **White women have more UI and SUI, prevalent and incident, than Black and Asian women,** possibly because of **higher urethral closure pressure in Black women.** The difference is less clear for OAB/UUI and against Hispanic women, and differences in MUI/UUI are probably **inconsequential.**
- **Rural residence carries more risk than urban. White and Black women seek care equally,** modified by leakage severity. In the National Health and Nutrition Examination Survey (NHANES), **lower socioeconomic status carries the greatest risk of UUI.**
- **Cross-cultural differences are unclear;** worldwide prevalence in older women is ≈28%, with marked heterogeneity.
- **Fig. 108.3** — United States prevalence of OAB, UUI, SUI and POP by race and ethnicity (White, Black, Hispanic, Asian, other).

Hormonal Therapy

ORAL VS VAGINAL OESTROGEN AND UI

Oral oestrogen ± progestogen	Topical (vaginal) oestrogen
Associated with incident UI in middle-aged and older women. In a large, well-conducted placebo-controlled trial in postmenopausal women, incident UI nearly doubled within 1 year	Not clearly linked to SUI. Proven efficacy for vaginal atrophy and recurrent urinary tract infection (UTI)

Obesity

- **The presence, duration and severity of UI are strongly associated with obesity,** most strongly for **SUI and MUI,** but for all types.
- **BMI ≥30 kg/m² more than doubles UI risk. Overweight without obesity already raises risk,** so the effect of weight gain is progressive. **Visceral fat, metabolic syndrome** (obesity, dyslipidaemia, raised blood pressure and fasting glucose) and **years of obesity** all add to likelihood and severity.
- **Weight-related UI is reversible.** Surgical or structured weight loss improves UI for as long as the weight stays off. Non-surgical programmes (diet, exercise, drugs) give a **significant but modest fall in daily episodes** (meta-analysis).
- **Obesity has little effect on the success of SUI procedures, but may reduce that of third-line OAB/UUI treatments.**

Smoking

- **The data are inconsistent, but consensus is growing for a link, strongest with urgency.**
- **Current smoking → urgency and frequency, dose-dependently** (heavy > light). **Waterpipe smoking** is also linked to OAB. **UUI is linked especially in younger women.** Severe UI is associated with current smoking in a large nurses' cohort, and smokers awaiting SUI surgery have more episodes.

Diet

- **Caffeine, especially coffee, has the most compelling link: UUI, MUI and OAB,** in both sexes. **Diet has no consistent link with SUI.** Carbonated drinks and artificial sweeteners are linked mainly to urgency but remain unconfirmed.

Medical Conditions

- **Diabetes mellitus (DM) and depression are the principal medical associations.**
- **Type 2 DM: up to twice the UI prevalence** of age-matched non-diabetic women, with nearly half of diabetic nurses reporting at least monthly UI. **Obesity amplifies the risk. Macroalbuminuria and peripheral neuropathic pain are independent risk factors** (NHANES). Type 1 DM data are emerging but less consistent.
- **Depression is linked multifactorially and bidirectionally:** UI lowers perceived self-health, and a shared pathophysiology is proposed. **Depression raises the risk of concurrent and later UI.**
- **Multimorbidity:** hypertension, osteoarthritis and heart failure are each independently associated, and their cumulative effect can tip vulnerable women into UI.

KEY POINTS: RISK FACTORS FOR URINARY INCONTINENCE IN FEMALES

- **Age:** strongest non-modifiable factor; **LTCF residence** independent of age. **Delivery:** CS advantage lost with one vaginal birth. **Parity:** each delivery adds risk, mainly SUI.
- **Oral oestrogen** → SUI; topical does not. **Obesity:** dose- and duration-dependent, reversible. **Smoking:** urgency and UUI; data on stopping are scant. **Caffeine:** OAB, not SUI. **DM and depression:** commonest medical associations.

Concept — *The risk factors split into two groups. **Obstetric injury and loss of support produce stress incontinence** (vaginal delivery causes SUI, not UUI). **Ageing, neurological, metabolic and lifestyle factors** (smoking, caffeine, DM, depression, immobility) **push towards urgency**. This is why the dominant type of UI moves from stress to urge across a woman's lifetime.*

EPIDEMIOLOGY OF URINARY INCONTINENCE IN MALES

Prevalence, Incidence, Remission Rates

- **UI is considerably less prevalent in men. SUI is uncommon except after radical pelvic surgery** (radical prostatectomy (RP), abdominoperineal resection), after surgery for benign prostatic enlargement, or with neurological disease.
- **UUI and MUI are commoner than SUI**, often from **BOO due to benign prostatic enlargement**, other outlet disorders, OAB, or inflammation and infection.
- **Prevalence in older men: 11–34%; up to 11% have daily UI. Incidence depends on age: ≈1–10% per year at 60–70.**
- **Remission is higher than in women, up to 40%**, because male leakage is mostly urgency-related and often has reversible (infective or inflammatory) causes.

Risk Factors for Urinary Incontinence in Males

- **Age is the strongest independent risk factor, possibly more so than in women.** Other factors: **infection** (acute cystitis, prostatitis), which also rises with age; **obesity indicators; loss of grip and quadriceps strength; possibly chronic obstructive pulmonary disease (COPD).** **Non-Hispanic Black men have more UUI** than Hispanic and non-Hispanic White men.
- **RP, radical pelvic surgery and radiotherapy:** UI 1 year after RP is **8–60%, depending mainly on the definition used.**
- **Less UI after RP with:** no pre-existing LUTS or previous transurethral resection of the prostate (TURP); **bladder-neck, neurovascular-bundle and possibly Retzius sparing;** younger age. The data conflict.

- **Robotic vs open RP: comparable UI.** Robotic vs laparoscopic: earlier and better continence at 1 year. **RP vs external-beam radiotherapy with low-dose-rate brachytherapy:** RP gives more UI at 5 years but fewer OAB and bowel symptoms.
- **TURP (electrosurgical or potassium titanyl phosphate (KTP) laser): UI ≈1%.**

UI EPIDEMIOLOGY — WOMEN VS MEN

	Women	Men
Prevalence	Higher	Considerably lower; 11–34% in older men
Commonest type	SUI when younger → MUI/UUI when older	UUI > MUI > SUI; in some studies of elderly men, UUI exceeds the female rate
Cause of SUI	Obstetric injury, loss of support, intrinsic sphincteric deficiency (ISD)	Iatrogenic (RP, radical pelvic surgery) or neurological
Age effect	Independent risk factor	Even stronger
Remission	Up to ≈12% per year; lowest for MUI	Up to 40% (reversible urgency causes)

KEY POINTS: EPIDEMIOLOGIC ASPECTS OF URINARY INCONTINENCE IN MALES

- **UUI > MUI > SUI;** SUI follows radical pelvic surgery (especially RP) or neurological disease. SUI after RP **8–60%**; remission up to **40%**.

Concept — Men have a **fixed, powerful bladder neck and sphincter**, so male UI is **mostly urgency-driven and often reversible**, and male SUI is **almost always iatrogenic**. This explains both the low prevalence and the high remission rate.

DEFINITIONS AND CLASSIFICATION OF PELVIC ORGAN PROLAPSE

- **POP is the downward displacement of the pelvic organs**, with protrusion of the uterus and/or vaginal compartments and the organs around them (bladder, rectum, small or large bowel). It results from **loss of support of one or more vaginal compartments**.
- **Describe POP by vaginal segment, not by organ.** Avoid **cystocele, rectocele, enterocele** and **urethrovesical junction prolapse**, because the structure behind the bulge is uncertain, especially after previous POP surgery.

COMPARTMENTS OF POP

Compartment	What descends	Contents and notes
Anterior	Anterior vaginal wall	Usually bladder; an anterior enterocele is possible after reconstructive surgery. The commonest segment to prolapse
Apical	Uterus/cervix, or the vaginal cuff after hysterectomy	May include small bowel. Enterocele: a peritoneal sac containing bowel ± omentum descending into the rectovaginal space. It may be asymptomatic or cause defecatory dysfunction and pelvic pain or pressure
Posterior	Posterior vaginal wall	Rectum, but also small bowel or colon, even with the uterus in place
Procidentia	Total vaginal eversion	Complete uterine or vaginal-cuff prolapse

Symptoms of Pelvic Organ Prolapse

- **Most specific symptom: a vaginal bulge the patient can see or feel.** Others: pelvic pressure or discomfort, discharge, **splinting or digitation** to void or defecate, low backache.

- **Urinary and bowel symptoms correlate poorly with POP site and severity.** Frequency, nocturia, UI, faecal incontinence (FI), voiding and defecatory dysfunction and incomplete emptying should still be assessed, but they often reflect **coexisting bladder, anorectal or pelvic floor dysfunction**. Bowel symptoms correlate weakly at most.
- **Constipation without a bulge should rarely prompt posterior repair.** Exception: **stool caught "in a pocket" needing splinting**, which suggests a rectocele defect.
- **Bulge and splinting correlate much better once the leading edge is beyond the hymen.**
- **Establish when SUI began.** SUI falls in prevalence and severity as POP passes the hymen, possibly because urethral kinking obstructs the urethra, so concomitant SUI treatment may be indicated in some women.
- **Sexual function:** POP harms body image and sexual function, and treatment can improve both. **Ask sexually inactive women whether they wish to resume**, because this affects the choice of repair.
- **Never attribute abnormal or postmenopausal bleeding to POP;** it needs gynaecological evaluation before any POP intervention.
- **There is little evidence that POP causes pain**, so look for another source first.
- **A pessary trial is diagnostic:** if symptoms improve, POP is the likely cause and **surgery is more likely to help**.

Physical Examination of Pelvic Organ Prolapse

BADEN-WALKER HALFWAY SYSTEM VS PELVIC ORGAN PROLAPSE QUANTIFICATION (POP-Q)

	Baden-Walker	POP-Q
Reference point	Hymen	Hymen (plane of hymen = 0)
Sites	Six midline sites: urethral, vesical, uterine, cul-de-sac, rectal, perineal	Nine measurements: six vaginal points (Aa, Ba, C; Ap, Bp, D) and three landmarks (gh, pb, tvl)
Scale	Grades 0–4: 0 none; 1 halfway to the hymen; 2 to the hymen; 3 halfway past the hymen; 4 maximum	Centimetres → stages 0–4
Weakness or strength	No exact location of specific structures; high interobserver variability	Precise and objective; better interobserver agreement and reliability
Status	One of the most commonly used	Developed 1993, published 1996 (Bump), updated 2002; the recommended staging system (IUGA/ICS 2016) for both research and routine clinical use

POP-Q POINTS AND LANDMARKS

Point	Definition	Range or value	Measured
Aa	Anterior wall, 3 cm proximal to the hymen ≈ urethrovesical junction	–3 to +3; –3 = no prolapse	Maximal strain
Ba	Most distal point of the anterior wall between Aa and the apex	–3 if no prolapse; equals C in total uterine prolapse or post-hysterectomy eversion	Maximal strain
C	Most distal edge of the cervix, or leading edge of the vaginal cuff after hysterectomy	—	Maximal strain
D	Posterior fornix/pouch of Douglas: level of uterosacral ligament attachment to the proximal posterior cervix	Omitted if there is no cervix	Maximal strain

Point	Definition	Range or value	Measured
Ap	Posterior wall, 3 cm proximal to the hymen (parallels Aa)	-3 to +3	Maximal strain
Bp	Most distal point of the posterior wall between Ap and the apex	-3 if no prolapse; equals C in total prolapse	Maximal strain
gh	Genital hiatus: middle of the urethral meatus → posterior hymenal ring	—	Rest
pb	Perineal body: posterior margin of the genital hiatus → mid-anal opening	—	Rest
tvI	Total vaginal length: posterior fornix → hymen with C or D reduced to its normal position	—	Rest

- **Convention:** all values are in cm. **Negative values lie inside the introitus and positive values beyond it.** Maximal prolapse is shown by straining or Valsalva, supine or standing.
- **The results are recorded on a 3 × 3 grid** (Fig. 108.5): Aa, Ba, C / gh, pb, tvI / Ap, Bp, D.
- **Measurements taken straight after pessary removal are unreliable and underestimate prolapse.**

POP-Q STAGING (BY THE MOST DISTAL PORTION OF THE PROLAPSE ON MAXIMAL STRAIN)

Stage	Criterion
0	No prolapse: Aa, Ba, Ap and Bp all -3; C or D between -tvI and -(tvI - 2) cm
1	Most distal portion >1 cm above the hymen (< -1)
2	Most distal portion between 1 cm above and 1 cm below the hymen (-1 to +1)
3	>1 cm beyond the hymen but not the whole vagina: > +1 and < +(tvI - 2) cm
4	Complete eversion, or eversion to within 2 cm of tvI: ≥ +(tvI - 2) cm

- **Worked examples (Fig. 108.6).** Post-hysterectomy total vault eversion: Aa +3, Ba +8, C +8; gh 4.5, pb 1.5, tvI 8; Ap +3, Bp +8 → stage 4. Normal support: Aa -3, Ba -3, C -8; gh 2, pb 3, tvI 10; Ap -3, Bp -3, D -10.
- **Simplified POP-Q:** the same ordinal staging, but **only four points (Ba, Bp, C, and D if the uterus is present)**, measured in dorsal lithotomy with Valsalva.
- **Traps:** stage is set by the single most distal point. **Aa and Ap cannot exceed +3. Ba and Bp equal C in total eversion. There is no D after hysterectomy.** Pessary removal falsely lowers the measurements.

Imaging

- **History plus POP-Q examination is usually enough; imaging is generally unnecessary.**
- **Pelvic ultrasound** identifies pelvic organ abnormalities and **mesh from previous POP or UI surgery.** Ultrasound assessment of the anatomical defect is **investigational.**
- **Defecography ± dynamic magnetic resonance imaging (MRI)** is used for incomplete bowel emptying, significant defecatory dysfunction, or **examination-symptom mismatch.** Rest and straining MRI is **particularly useful for enterocele** (Fig. 108.7: A, severe cystocele with uterine descent; B, defecatory dysfunction with severe enterocele).
- **Routine preoperative MRI has not been shown to be necessary, cost-effective or outcome-improving,** so it is discretionary. **Careful POP-Q examination, reassessed at surgery if needed, remains the cornerstone.**

KEY POINTS: DEFINITION AND CLASSIFICATION OF PELVIC ORGAN PROLAPSE

- Name the **vaginal segment**, not the organ. **Stage with POP-Q**; careful POP-Q examination is the cornerstone of assessment.

Concept — POP-Q is reproducible because *every point is measured in centimetres from one fixed landmark, the hymen*. For the same reason, prolapse is described by *the vaginal segment that moves, not by the organ presumed to lie behind it*.

EPIDEMIOLOGY OF PELVIC ORGAN PROLAPSE

- **Prevalence is hard to estimate.** Studies use symptoms or examination, rarely both. Definitions vary because **normal and abnormal support are poorly defined**, and symptoms correlate poorly with POP location and severity.
- **Seeing or feeling a bulge is the symptom most strongly correlated with POP beyond the hymen**, and POP is often asymptomatic until it reaches the hymenal ring.
- **Some loss of support is found on examination in 43–90% of women.**
- **Stage 1 anterior and posterior POP is probably normal.** In the Pelvic Organ Support Study (POSST) of ≈1000 women at routine gynaecological care, **only 24% had no POP**; stage 1 was found in 38%, stage 2 in 35% and stage 3 in 2%. **Asymptomatic stage 1–2 POP is therefore normal and rarely needs treatment.**
- **Women's Health Initiative (WHI)** (postmenopausal women, examined): any POP in **41% with a uterus and 38% after hysterectomy**; anterior 34%, posterior 19%, uterine 14%.
- **By compartment: anterior > posterior > apical.** However, **apical POP is under-diagnosed**, because it almost always accompanies significant anterior or posterior POP, especially anterior. Significant apical descent ($C \geq -3$) becomes more common with each stage of anterior POP and is **universal at stage 4**.
- **When planning a cystocele repair, suspect apical POP and consider a concomitant vault suspension.**
- **Symptomatic POP** (sensation of bulge, pressure or protrusion, or a visible bulge): **3–12%.** **Questionnaires under-report compared with examination:** only about a fifth of women with stage ≥ 2 report symptoms, partly from shame and reluctance to discuss it.
- **Bother varies:** at **stage 2**, Hispanic and Native American women report more bother than non-Hispanic White women; at **stage ≥ 3** there is no difference.

Natural history

- **WHI Estrogen Plus Progestin Trial subgroup** (2–8 years): annual incidence per 100 woman-years of cystocele 9.3, rectocele 5.7, uterine prolapse 1.5. **Spontaneous regression was common, especially for grade 1.**
- **Symptomatic women choosing observation mostly stay unchanged**; a minority progress over ≈1–1.5 years and very few regress.
- **Progression risk:** age >60 , Ba ≥ 4 cm, genital hiatus ≥ 5 cm at baseline. **Change in gh over time**, not only its size, predicts new POP.
- **Natural history: very slow progression, if any**, even in symptomatic advanced POP.

Risk Factors for Pelvic Organ Prolapse

- **Multifactorial; best established: vaginal childbirth, age, obesity.**
- **Vaginal childbirth is the commonest association.** Oxford Family Planning Study (prospective, ≈17,000 women): **parity was the strongest factor**, with adjusted RR 8.4 for two deliveries and 10.85 for ≥ 4 . WHI:

one birth roughly doubles the odds of uterine prolapse, cystocele and rectocele (OR ≈ 1.9 – 2.2), and **each further delivery up to five adds 10–20%**.

- **Forceps raise the risk further**, including the risk of POP surgery. **Prevalence gradient: nulliparous < CS only < spontaneous or vacuum vaginal < forceps.**
- **CS protects, but by an unclear amount. Elective (unlaboured) CS $\approx$ nulliparous; CS after labour $\approx$ vaginal delivery.** Routine elective CS was calculated to prevent 46% of POP in one population.
- **Age:** relative prevalence rises $\approx 40\%$ per decade (POSST), and WHI women aged 60–79 had more POP than those aged 50–59. **Surgery for POP and UI peaks in the seventh decade**, and is more likely if the **first delivery was before 25**.
- **Obesity:** in the WHI, overweight and obesity raised uterine, posterior and anterior POP, and **waist ≥ 88 cm** raised rectocele and cystocele. In POSST, **BMI 25–30 and >30 each gave ≈ 2.5 times the odds**. Meta-analysis: **RR ≥ 1.36 for overweight, ≥ 1.47 for obesity.**
- **Hysterectomy raises the risk, most when done for POP.** Oxford study: POP surgery **29 vs 16 per 1000 woman-years**; cumulative **1% at 3 years $\rightarrow$ 5% at 15 years**; **5.5 times higher if the hysterectomy was for POP**. This strongly predicts repeat surgery, partly through **omission of a concomitant vault suspension**.
- **Race:** Black women lowest, Hispanic highest after adjustment. WHI: **Hispanic and Asian women more cystocele, Black women less**; symptom data agree.
- **Other factors:** maximum birth weight (inconsistent). **Menopause is often cited, but oestrogen status is not consistently associated.** **Genetics:** POP in a mother or sister raises risk, **≈ 5 times in siblings of women with severe POP**, though some studies disagree. **Chronic constipation gives 2.5 times the risk of symptomatic POP.** Smoking, heavy lifting, intense activity and COPD are inconsistent.
- **Recurrence after surgery: preoperative stage 3–4.**

SHARED RISK FACTORS — URINARY INCONTINENCE (WOMEN) VS PELVIC ORGAN PROLAPSE

Factor	UI	POP
Vaginal delivery	Leads to SUI, not UUI; CS protects only if every delivery is CS ; NNT 13	Strongest factor ; each birth adds risk; CS after labour $\approx$ vaginal delivery
Forceps	Suggested, not conclusive	Increases risk
Birth weight	Consistent (≥ 4 kg)	Inconsistent
Age at first delivery	Conflicting (near 20 vs >30)	<25 raises the risk of surgery
Ageing	Independent; type shifts to UUI/MUI	$\approx 40\%$ rise per decade; surgery peaks in the 7th decade
Obesity	BMI ≥ 30 more than doubles risk; reversible	≈ 1.4 – 2.5 -fold
Race	White > Black/Asian	Hispanic highest, Black lowest
Oestrogen	Oral oestrogen increases UI ; vaginal does not	Not consistently associated
Hysterectomy	Both de novo UI and resolution	Increases risk, especially if done for POP
Constipation	Risk factor for SUI in pregnancy	2.5-fold risk of symptomatic POP
Smoking	Urgency and UUI	Inconsistent

- **What the table shows:** vaginal birth, age and obesity are shared by UI and POP. **Race runs in opposite directions for Hispanic women, and oestrogen has opposite significance.** These are the usual examination traps.

KEY POINTS: EPIDEMIOLOGY OF PELVIC ORGAN PROLAPSE

- Symptomatic POP **3–12%**; asymptomatic POP in most adult women. **Anterior is commonest**. Risk rises with age, each vaginal birth, **forceps**, obesity and hysterectomy (especially for POP); **elective CS protects**. White and Hispanic > Black.

Concept — POP is largely a birth-injury disorder, made manifest by age and load. Labour, not the route of delivery alone, causes the damage. Stage 1–2 is near-universal and progression is slow, so the trigger for treatment is symptoms (a bulge at or beyond the hymen), not anatomy.

RELATIONSHIP BETWEEN URINARY INCONTINENCE AND PELVIC ORGAN PROLAPSE

- UUI, SUI and POP share aetiological risk factors. 40–55% of women with SUI have significant anterior POP; ≈20% of women with OAB have anterior POP.

POP and OAB

- **OAB is commoner with POP**. Frequency tracks anterior and apical POP; UUI tracks the severity of anterior POP.
- **OAB improves after POP surgery in 60–80%**. Anterior repair reduces frequency, UUI and voiding difficulty, and **anterior or apical repair helps OAB more than posterior repair**. **No reliable predictor exists** of whether OAB will persist.
- **Mechanisms (unclear): BOO is probably the most important**. Chronic bladder-wall stretch may cause DO. A severe cystocele may pull the bladder neck open, so urine enters the proximal urethra and triggers DO.

POP and SUI

- **More than half of women with SUI have POP**. In the **Trial of Mid-Urethral Slings (TOMUS)** (National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK); two mid-urethral sling (MUS) types): **stage 2 POP in 49%, stage 3 in 6%. 78% with stage 3 were symptomatic vs 25% with stage 2**, and 15% of asymptomatic stage 2 women had concomitant repair.
- **Missing POP at UI surgery increases later POP surgery**. In Medicare claims, urologists did concomitant POP repair at sling surgery about half as often as gynaecologists, and **their patients had ≈twice the 12-month rate of repeat prolapse surgery**. **Identify and manage symptomatic POP when evaluating UI**.
- **Occult SUI is SUI seen only after the coexisting POP is reduced**, in up to a quarter of advanced POP. De novo SUI after POP surgery is more likely with **age >66, DM, or low urethral closure pressure** on UDS.
- **Sequence: cystocele → urethral kink → obstruction masks SUI → repair relieves obstruction → SUI unmasked**. Conversely, SUI often improves as POP worsens. **UDS with the prolapse reduced** identifies occult SUI.

TRIALS OF A CONCOMITANT CONTINENCE PROCEDURE AT PROLAPSE SURGERY

Trial	Population	Intervention vs comparator	Result	Verdict
CARE (Colpopexy and Urinary Reduction Efforts), subgroup	Stress-continent women, stage 2–4 POP, abdominal sacrocolpopexy; UDS with the prolapse reduced	Burch colposuspension vs none	Postoperative SUI in those with SUI on reduction UDS: 58% without Burch vs 38% with (P = 0.04)	SUI on reduction UDS predicts postoperative SUI

Trial	Population	Intervention vs comparator	Result	Verdict
OPUS (Outcomes Following Vaginal Prolapse Repair and Midurethral Sling)	Women without SUI symptoms having vaginal POP surgery (randomised)	Prophylactic MUS vs sham	UI at 1 year 27.3% vs 43% (P = 0.002) ; NNT 6.3	Fewer de novo SUI, but more adverse events (below)

- **OPUS harms with a sling vs sham** (all $P \leq 0.05$): bladder perforation **6.7% vs 0**, UTI **31.0% vs 18.3%**, major bleeding **3.1% vs 0**, incomplete emptying at 6 weeks **3.7% vs 0**.
- **What the table shows:** a prophylactic continence procedure prevents de novo SUI (CARE for the abdominal route, OPUS for the vaginal route), but at a cost. **Counselling before POP surgery must cover both the benefit and the added risk of a prophylactic sling.**

KEY POINTS: CONDITIONS ASSOCIATED WITH PELVIC ORGAN PROLAPSE

- **>40% of SUI and 20% of OAB** have anterior POP; repair improves OAB. **Unaddressed occult SUI** may be worse after POP surgery.

Concept — Anterior prolapse and SUI are mechanically coupled. The same cystocele that kinks the urethra both **masks stress leakage and drives urgency**, so one repair can relieve OAB and unmask SUI at the same time.

CONSEQUENCES OF URINARY INCONTINENCE AND PROLAPSE

Societal Costs of Urinary Incontinence

- **Costs are hard to measure**, because they include **out-of-pocket items payers do not capture** (pads, laundry, missed work) as well as tests, visits, drugs, surgery, and treatment of UTI and skin breakdown.
- **Female SUI alone: >\$12 billion a year in the United States, >70% out-of-pocket.** Severe UI carries substantial out-of-pocket care costs, which fall after surgery (Urinary Incontinence Treatment Network (UITN)). SUI/MUI means more visits and spending than in controls.
- **OAB: ≈42 million Americans, ≈\$25 billion a year**, with substantially higher direct and indirect costs than in people without OAB.

Social Impact of Urinary Incontinence

- **UI affects nearly every aspect of life.** Relationships and sex suffer, and **surgery for UI improves sexual function**. Patients avoid travel, **restrict food and fluid (sometimes to inappropriate dehydration)**, wear dark clothing and avoid strangers. Work productivity falls, and **sleep disturbance** reduces daytime alertness.
- **UI affects all ages.** Young nulligravid women with UI report much lower well-being. In the elderly, **severity rather than type determines the QoL impact**. QoL declines progressively over 10 years in both sexes, across nearly all domains, **possibly more than with arthritis or DM**.
- **The most bothersome LUTS are urgency and UUI.**

KEY POINTS: SOCIAL CONSEQUENCES OF URINARY INCONTINENCE

- **>\$50 billion a year** in direct and indirect costs of UUI plus SUI in the United States. Relationships, sex, travel, work, diet and sleep are all affected.

Impact of Pelvic Organ Prolapse on Quality of Life

- **POP significantly reduces QoL**; shame makes women reluctant to discuss it, even with their physician. **Body image is an important determinant of QoL in POP.** POP costs measurable disability-adjusted life-years (DALYs) in middle-aged women, and **surgery improves QoL.**
- **≈38% of women with POP avoid intercourse**, and **POP plus UI** reduces libido, excitement and orgasm more than UI alone.

Financial Impact of Pelvic Organ Prolapse

- **Lifetime risk of one operation for POP or UI: 11% (Olsen, 1997); reoperation 29%.** The lifetime risk of POP surgery in Western Australia is 19%.
- **POP is the commonest indication for hysterectomy in American women over 54.**
- **1997 direct costs of POP surgery were ≈\$1 billion:** vaginal hysterectomy ≈half, cystocele/rectocele repair ≈28%, abdominal hysterectomy ≈13%. A UI procedure was included in 21% of POP operations.
- **Demand will rise with ageing.** Women with ≥1 pelvic floor disorder are projected to increase from **28.1 million (2010) to 43.8 million (2050)**; POP +46%; POP surgery +48% (166,000 → ≈246,000).

Concept — UI and POP are *non-lethal conditions whose burden is carried in quality of life and out-of-pocket cost.* That is why bother, not anatomy, triggers treatment, and why an ageing female population will drive demand.

PHYSIOLOGY OF URINARY CONTINENCE

Overview of Normal Continence Mechanisms

- **Continence is maintained by an interplay of neural, structural and ultrastructural mechanisms** in the LUT and the structures around it. A change in any one can cause UI, and diseases, their treatments and ageing often affect several at once.
- **Example — DM:** hyperglycaemic polyuria first → microvascular change and oxidative stress → direct neurogenic and myogenic damage → UUI and SUI.

Neural Control of the Lower Urinary Tract

NEURAL CONTROL DURING BLADDER STORAGE

Pathway	Nerve or centre	State and effect during filling
Parasympathetic	Pelvic nerve	Suppressed , so no detrusor contraction
Sympathetic	Hypogastric nerve	Active: β-mediated detrusor relaxation and α-mediated bladder-neck closure
Afferent	Aδ myelinated fibres	A spinal reflex drives sympathetic bladder-neck closure, which is much stronger in men
Somatic	Pudendal nerve (Onuf's nucleus, sacral cord)	Continuous external-sphincter resistance
Central	Pontine micturition centre; suprapontine centres	The pons integrates afferent input and triggers parasympathetic contraction; suprapontine centres tonically inhibit it until voiding is appropriate

- **Parasympathetic suppression together with sympathetic activity creates the low-pressure reservoir**, and it is maintained throughout filling.

Bladder: An Organ Capable of Significant Expansion at Low Pressures

- **The bladder's position allows expansion while protecting the sphincter.** The dome expands into the peritoneal cavity with virtually no opposing force, while the base and neck sit in a stable retropubic position. **In men this position is fixed**, so leakage needs a direct injury to the sphincteric unit (or a neurological, traumatic or operative event). **In women, pelvic floor laxity alters bladder-neck position and function.**
- **Poor electrical coupling between detrusor smooth muscle cells** stops spontaneous single-cell contractions, triggered by stretch, from spreading into a coordinated contraction. A sustained contraction needs **coordinated parasympathetic drive.**
- **The multilayered urothelium stores urine efficiently:** its "umbrella cells" flatten with filling.
- **The extracellular matrix is critical to compliance**, in particular the collagen type (**type I in normally compliant bladders**) and the **collagen-to-elastin ratio**. Inflammation, infection and radiation alter compliance.
- **Tonic sympathetic, β -mediated detrusor relaxation** further lowers intravesical pressure.

KEY POINTS: FACTORS CONTRIBUTING TO LOW-PRESSURE RESERVOIR DURING BLADDER FILLING

- **Neural:** parasympathetic suppression and sympathetic activation. **Gross anatomy:** the intraperitoneal dome expands freely.
- **Microanatomy:** the multilayered mucosa expands and collapses; poor detrusor cell coupling dissipates aberrant contractions; the extracellular matrix maintains compliance.

Bladder Outlet and Sphincteric Mechanisms

- **Outlet resistance must exceed storage pressure;** otherwise a low-pressure reservoir is useless.
- **A competent outlet has three components:** the urethral mucosa, the **internal sphincter** (intrinsic properties of the bladder neck and proximal urethra) and the **distal rhabdosphincter.**
- **Longitudinal mucosal folds** let the urethra distend for voiding and form a **mucosal seal** during filling. The **vascular, spongy submucosa** adds distensibility.

Male Sphincteric Mechanisms

- **The internal sphincter runs from the bladder neck to the distal verumontanum,** and normally **urine does not pass beyond the bladder neck.** α -Adrenergic smooth muscle at the bladder neck and in the prostatic urethra extends towards the rhabdosphincter, where **the greatest intraurethral forces are generated.** **α -Activation here, with β -mediated relaxation of the bladder body, keeps the patient dry,** and the prostate adds resistance.
- **The external sphincter is concentric, mostly type I (slow-twitch) skeletal muscle with some smooth muscle.** The slow-twitch fibres give **tonic closure at >40 cm H₂O** throughout filling, rising with voluntary contraction. **Fast-twitch levator fibres** add transient closure under stress. **Pudendal injury or denervation** weakens the sphincter and causes UI.
- **The pubourethral ligaments (anteriorly) and the musculofascial plate fusing at the perineal body (posteriorly)** stabilise the sphincter against external forces.

Female Sphincteric Mechanisms

- **Continence arises mainly along the proximal and mid-urethra, and an incompetent bladder neck is common in continent women.** **Circular striated muscle** compresses the mid-urethra almost circumferentially under **tonic pudendal stimulation.** In rats, striated fibres run transversely and

longitudinally and insert into the urethral wall. **Pudendal denervation from prolonged labour** is one route from delivery to UI.

- **Distally, the striated fibres lie ventrally** as the **compressor urethrae** (from the perineal membrane) and the **urethrovaginal sphincter** (from the vaginal wall).
- **Seal:** the spongy, **oestrogen-sensitive submucosa** brings the mucosal surfaces together and provides **up to 30% of continence forces**.
- **Support:** the **anterior vaginal wall** is a firm backing for mid-urethral compression. The **pubourethral ligaments** limit proximal movement and keep abdominal force off the rest of the urethra. The **urethropelvic ligaments** anchor the urethra to the arcus tendineus. **Lax vaginal support with intact pubourethral ligaments causes shear** in the continence zone.
- **Continence needs support, intrinsic urethral properties and muscle together;** losing one, and usually several, causes UI.

MALE VS FEMALE SPHINCTERIC MECHANISMS

	Male	Female
Main continence zone	Bladder neck to verumontanum, plus the rhabdosphincter	Proximal and mid-urethra
Bladder neck	Powerful (α -adrenergic, plus prostate)	Weak; often incompetent
External sphincter	Slow-twitch rhabdosphincter	Mid-urethral circular striated muscle
Loss of one unit	Tolerated: bladder neck lost at TURP → intact rhabdosphincter keeps continence; external sphincter lost (pelvic trauma) → intact bladder neck keeps continence	Not tolerated: any external-sphincter deficiency → UI
Exposure	Fixed and protected	Vulnerable to external forces

Concept — *The male outlet has two independent, protected sphincteric units, so either one alone can keep a man dry. The female outlet relies on a single mid-urethral sphincter compressed against a mobile backing, so female continence depends on support as much as on muscle.*

PATHOPHYSIOLOGY OF URINARY INCONTINENCE: GENERAL PRINCIPLES

Factors Affecting Bladder Storage

BLADDER CAUSES OF UI

Mechanism	Causes	Result
Loss of suprapontine inhibition	Cerebrovascular accident (CVA), multiple sclerosis (MS), Parkinson disease; DM, even early in its course	Neurogenic detrusor overactivity (NDO) → UUI
Outlet obstruction	Women: obstruction after anti-incontinence surgery. Men: BOO from prostatic enlargement	DO → de novo UUI
Detrusor underactivity or areflexia	Lumbosacral cord or conus disease; peripheral neuropathy (DM, tabes dorsalis, alcoholism); radical pelvic surgery (radical hysterectomy, abdominoperineal resection), sometimes permanent; narcotics, including tramadol	Retention → overflow UI
Reduced compliance	Pelvic external-beam radiotherapy	Raised detrusor leak point pressure → UI

- **DM acts in two phases:** early UUI → later loss of sensation and contractility → poor emptying, UTI and UI.

Factors Affecting Sphincteric Function

- **ISD is mostly iatrogenic, especially in men;** less often neurological (e.g. **traumatic or vascular lumbosacral cord injury**).
- **Multiple system atrophy** destroys intrinsic sphincter function in men, with **≥20% UI after TURP**, which matters when counselling.
- **Cervical or upper thoracic spinal cord injury (SCI) → detrusor–sphincter dyssynergia (DSD) with NDO** → poor emptying and UUI, as sustained contractions overcome the outlet.
- **Drugs: α-blockers**, and any drug with α-antagonist or skeletal-muscle-relaxant action, can lower outlet resistance enough to cause leakage.
- **RP is the commonest surgical cause of male UI**, by direct sphincter damage. On UDS in incontinent men, **ISD predominates (>70%)**; DO and altered compliance are less common causes but contribute in >30%. **DO is common after RP whether or not the man is incontinent.** Functional urethral length and **maximum urethral closure pressure (MUCP)** fall.
- **Prior TURP or enucleation does not appear to raise post-prostatectomy incontinence (PPI)** in high-volume centres (controversial). **Bladder-neck preservation, Retzius sparing and nerve sparing** speed recovery and lower PPI.
- **Women:** urethral or anti-incontinence surgery can cause **scarring, periurethral fibrosis and ISD**, with risk rising with each failed operation. **Advanced POP surgery without outlet treatment increases postoperative UI** (abdominal sacrocolpopexy and vaginal repair of significant anterior POP), weighed against **overtreating women who would never develop significant UI**.
- **Labour and delivery:** prolonged labour, **third-degree lacerations**, large birth weight, multiparity and forceps are linked to sphincter dysfunction. Mechanisms: direct ligamentous or fascial injury, **ischaemia from prolonged compression**, pelvic or pudendal **denervation**, and direct genitourinary injury. Compared with CS, vaginal delivery is associated with more MUI, SUI, UUI and severe UI for up to 20 years.

Concept — Every UI is **either a bladder that fails to store** (overactivity, poor compliance, overflow) **or an outlet that fails to resist** (ISD, loss of support). A single disease or operation, such as DM, RP or pelvic radiotherapy, can do both.

PATHOPHYSIOLOGY OF STRESS URINARY INCONTINENCE IN FEMALES

- **Early theories focused on descent of the proximal urethra and bladder neck** out of their intra-abdominal location with anterior POP.

Loss of Urethral Support

- **Urethral support comes from the endopelvic fascia**, reinforced anteriorly by the pubourethral ligament and posteriorly by the vaginal wall. Together these normally prevent excessive urethral mobility.
- **Three theories build on this** (see the table below): the **hammock hypothesis** (backing), the **integral theory** (ligamentous laxity) and **ISD** (urethral function).

Intrinsic Sphincteric Deficiency

- **Support does not explain everything.** Some women with extensive relaxation and hypermobility are continent, while some with no hypermobility have severe SUI, typically those with a **scarred urethra or neurogenic disease**.
- **McGuire and Lytton (1978)** introduced ISD in their landmark description of the **autologous fascial pubovaginal sling** for urodynamically severe urethral dysfunction, usually after previous urethral surgery. Most patients improved, with no apparent obstruction on early assessment.

- **Blaivas and Olsson (1988) classified ISD as type III UI**, to distinguish it from mobility-related types. ISD means sphincter dysfunction, whether neural or structural.
- **"Pipestem" urethra**: a fixed urethra with little intrinsic closure, usually iatrogenic. **Subtler ISD coexists with hypermobility and probably underlies most SUI**. It may follow ischaemic injury (birth or other trauma) or progressive pudendal nerve damage.
- **Valsalva leak point pressure (VLPP)** is the abdominal pressure that causes urethral leakage. It **cannot be measured during a detrusor contraction or with altered compliance**. A **low VLPP (<60 cm H₂O)** indicates ISD, and historically directed patients to bulking agents or a pubovaginal sling.
- **Most female SUI involves some ISD**, even with hypermobility. This is why **pubovaginal sling, MUS and even bulking work in hypermobile patients**. **The converse is not true: Burch colposuspension**, which corrects hypermobility, is less effective with severe ISD and limited mobility.
- DeLancey emphasises urethral function by noting that **the MUS, the most studied support procedure, achieves only about 40% subjective success**.

THEORIES OF FEMALE SUI

Theory	Origin	Mechanism	Treatment implication
Pressure transmission	Enhorning, 1961	The hypermobile urethra descends out of the abdominal pressure zone, so abdominal force no longer compresses it; the proximal urethra becomes functionally part of the bladder, and the distal mechanisms cannot cope	Restore urethral position
Urethral hanging	Recent development of the above	The proximal urethra hangs below a less mobile bladder neck and pulls the bladder neck open (funnelling)	—
Hammock hypothesis	DeLancey, 1994	The urethra is not truly intra-abdominal. It is compressed against a firm backing : the anterior vaginal wall and its lateral attachments to levator ani and the arcus tendineus fasciae pelvis, together with the active mid-urethral sphincter. Loss of backing → SUI, worse if anterior support is intact (shear) and sphincter compression is weak	Restore posterior support; repositioning is not essential
Integral theory	Petros and Ulmsten, 1990	Maximal continence zone = mid-urethra at the pubourethral ligaments (PULs) . Pubococcygeus pulls the vagina forward against the PULs; backward forces close the proximal urethra. Connective-tissue (collagen/elastin) laxity → SUI; stretch receptors fire prematurely → urgency and UUI	Tension-free MUS
ISD	McGuire and Lytton, 1978	Failure of urethral function itself , not of its support	Sphincter-directed treatment (sling, bulking)

- **The pattern across the theories**: they move from **position** (pressure transmission, hanging), to **support** (hammock, integral), to **function** (ISD). **Trap**: the hammock hypothesis explicitly **rejects the idea that the urethra must be repositioned**.

Figure legends

- **Fig. 108.8** — Chronically raised intra-abdominal pressure (intravesical pressure = $P_{abd} + P_{det}$) **disrupts pubourethral support and funnels the bladder neck** (hanging → funnelling).

- **Fig. 108.10** — Lateral view with the urethra, vagina and fascia transected at the vesical neck: **downward abdominal force compresses the urethra against the supportive tissues.**

Concept — *Current thinking is that **almost all female SUI includes some ISD**. That is why **sphincter-directed procedures work in hypermobility but support-directed procedures fail in ISD**: the asymmetry follows from function outweighing position.*

PATHOPHYSIOLOGY OF INSENSIBLE INCONTINENCE

- **Even when loss is insensible, the usual causes are the common ones:** sphincteric deficiency or DOI. If UI cannot be demonstrated and the history is unclear, UDS and other tests help.
- **Consider rarer causes when standard non-operative treatment fails.**

LESS COMMON CAUSES OF INSENSIBLE INCONTINENCE

Cause	Typical pattern	Investigation
Urethral diverticulum	Post-void loss	Pelvic MRI
Ectopic ureter	Continuous loss	Upper tract imaging
Overflow UI	Frequent small-volume loss, frequency, small voids	UDS identifies it

- **Trap:** UDS may miss a diverticulum or an ectopic ureter, and is non-diagnostic fairly often.

PELVIC FLOOR SUPPORT MECHANISM

- The pelvic organs are supported by the endopelvic fascia, the perineal membrane, the levator ani complex and their attachments to the bony pelvis.
- The levator ani complex (pubococcygeus, puborectalis, iliococcygeus) is the primary support, forming a stable base on which the organs rest. The levator hiatus is the opening through which the urethra, vagina and rectum pass.
- The uterosacral and cardinal ligaments, which are condensations of the endopelvic fascia, hold the organs in the right position for the muscles to support them. POP follows disruption or dysfunction of any of these structures.

DELANCEY'S LEVELS OF VAGINAL SUPPORT (1992, CADAVERIC DISSECTION)

Level	Segment	Structures and attachment	Loss produces
I	Cervix and upper third of the vagina	Suspension to the sacrum and lateral sidewall. Parametrium (uterosacral-cardinal complex; uterus to sidewall) and paracolpium (to the vagina, with vertical fibres). The uterosacral ligaments are key	Uterine or vault prolapse. The vertical paracolpial fibres distinguish true eversion from post-hysterectomy cystocele, rectocele or enterocele with a well-suspended apex
II	Middle third	Lateral attachment: paravaginal attachment to the arcus tendineus fasciae pelvis (ATFP) and the superior fascia of levator ani; posteriorly, endopelvic fascia to the levator	Anterior vaginal wall POP: lateral (paravaginal) or central defect

Level	Segment	Structures and attachment	Loss produces
III	Distal third	Fusion: to the urethra anteriorly, the perineal body posteriorly and the levator ani laterally. Beneath lies the perineal membrane , attaching the urethra, vagina and perineal body to the ischiopubic rami and preventing rectal descent	Anterior loss → urethral hypermobility ; posterior loss → distal rectocele or perineal descent

- **ATFP:** a connective-tissue band running from the **lower sixth of the pubic bone, 1 cm lateral to the midline**, to just above the ischial spine.
- **Anterior wall defects, as a pair:** the **lateral defect** is paravaginal detachment from the ATFP; the **central defect** is a midline weakness of the perivesical fascia. **Richardson (1981)** found that the commonest defect at cystocele surgery was **ATFP detachment from the ischial spine**, and rarely from the pubis.
- **The level classification oversimplifies.** Loss at one level is rarely isolated, and **isolated cystoceles or rectoceles are rare because an apical component is almost always present.** Level I loss correlates strongly with anterior POP and moderately with posterior POP, so **identifying apical POP is crucial.**

Figure legend

- **Fig. 108.12** — The **cardinal and uterosacral ligaments support the cervix and, indirectly, the bladder base.** The retropubic, vesicovaginal and rectovaginal spaces are shown at the level of the cervix. (Modified from Raz, Stothers and Chopra, *Campbell's Urology*, 7th edition, 1998.)

Concept — Level I is the keystone. *Loss of apical support underlies most anterior and much posterior prolapse, so the apex must be assessed, and usually supported, in any repair.*

PATHOPHYSIOLOGY OF PELVIC ORGAN PROLAPSE

- **The urogenital hiatus** is the part of the levator hiatus ventral to the perineal body, and **all POP passes through it.** It is kept closed by **tonic levator activity**, which compresses the urethra, vagina and rectum against the pubis.
- **Levator trauma or denervation → loss of tone → widened urogenital hiatus → POP.** The hiatus widens further as POP progresses.
- **Childbirth is the commonest cause of levator trauma.** On MRI, **up to 20% of primiparous women have levator defects, which are not seen in nulliparous women.**
- **Complete levator avulsion** (found in about a quarter of symptomatic women on 3D/4D transperineal ultrasound) is associated with POP, **advanced POP (stage III–IV) and three-compartment POP.**
- **Muscle and fascia work together, as a sequence:** while the levator keeps the hiatus closed, the endopelvic fascia carries little load → once the muscle fails, the ligaments carry all of it → over time they fail → POP. Ligament injury or stretching occurs with **vaginal delivery, hysterectomy, chronic straining and ageing.**
- **Collagen metabolism is consistently altered in POP**, possibly with a changed **type I:III collagen ratio.** Whether this is cause or effect is unclear, but **Ehlers–Danlos and Marfan syndromes increase POP risk**, which supports a connective-tissue role.
- **Neuropathic injury.** The pelvic floor has two nerve supplies:
 - The **pudendal nerve** supplies the urethral and anal sphincters and the perineal muscles.
 - The **nerves to levator ani (S3–S5)** supply the main pelvic floor musculature.

Levator electromyography in primiparous women shows neuropathy in about a quarter at 6 weeks, rising slightly by 6 months. There is **almost none after elective CS**, but **similar injury after CS in labour.**

KEY POINTS: PELVIC FLOOR SUPPORT MECHANISM AND PATHOPHYSIOLOGY OF PELVIC ORGAN PROLAPSE

- Support comes from the endopelvic fascia, perineal membrane, levator ani and their bony attachments.
- **Level I** (uterus and upper vagina to the sacrum and sidewall) → apical prolapse. **Level II** (middle third) → anterior wall prolapse. **Level III** (distal) → urethral hypermobility, distal rectocele or perineal descent.
- POP is multifactorial: **direct muscle trauma, denervation, and injury or stretching of the endopelvic fascia.**

Concept — *In POP the muscle fails first and the fascia second. An intact levator unloads the ligaments. Once birth injury or denervation opens the hiatus, the ligaments carry the load alone and eventually fail. This is why labour, not caesarean delivery as such, is the injury.*